

Quiz (Habanero)

Q1. If $f(x) = x^2$ and $g(x) = 2x + 1$, find $(f + g)(x)$

- (A) $x^2 + 2x + 1$
- (B) $x^2 - 2x - 1$
- (C) $x^2 + 2x - 1$
- (D) $x^2 - 2x + 1$
- (E) $x^2 + x + 1$

Q2. The average temperature in a city is given by $T(x) = x + 5$ and the average rainfall is given by $R(x) = 2x - 3$. Find $(T - R)(x)$

- (A) $x + 8$
- (B) $x - 8$
- (C) $-x + 8$
- (D) $-x - 8$
- (E) $x + 3$

Q3. If $f(x) = x^2 - 4$ and $g(x) = x + 2$, find $(f \cdot g)(x)$

- (A) $x^3 - 2x^2 - 8$
- (B) $x^3 + 2x^2 - 8$
- (C) $x^3 - 2x^2 + 8$
- (D) $x^3 + 2x^2 + 8$
- (E) $x^3 - 4x^2 + 4$

Q4. The cost of maintaining a park is given by $C(x) = 2x + 10$ and the cost of landscaping is given by $L(x) = x^2 + 5$. Find $(C + L)(x)$

- (A) $x^2 + 2x + 15$
- (B) $x^2 + 2x + 10$
- (C) $x^2 + 2x - 5$
- (D) $x^2 - 2x + 10$
- (E) $x^2 + 2x + 5$

Q5. If $f(x) = x - 2$ and $g(x) = x^2 + 1$, find $(f \cdot g)(x)$

- (A) $x^3 - 2x^2 + x - 2$
- (B) $x^3 - 2x^2 - x + 2$
- (C) $x^3 + 2x^2 - x - 2$
- (D) $x^3 - 2x^2 - x - 2$
- (E) $x^3 - x^2 - x + 2$

Q6. The population of a species is given by $P(x) = x^2 + 3$ and the population of another species is given by $Q(x) = 2x - 1$. Find $(P - Q)(x)$

- (A) $x^2 - 2x + 4$
- (B) $x^2 - 2x - 4$
- (C) $x^2 + 2x + 4$
- (D) $x^2 + 2x - 4$
- (E) $x^2 - x + 4$

Q7. If $f(x) = x^2 - 1$ and $g(x) = x + 1$, find $(f \cdot g)(x)$

- (A) $x^3 + x^2 - x - 1$
- (B) $x^3 - x^2 - x + 1$
- (C) $x^3 + x^2 + x + 1$
- (D) $x^3 - x^2 + x - 1$
- (E) $x^3 - x^2 - x - 1$

Q8. The water level in a reservoir is given by $W(x) = 2x + 5$ and the water level in another reservoir is given by $V(x) = x^2 - 2$. Find $(W + V)(x)$

- (A) $x^2 + 2x + 3$
- (B) $x^2 + 2x + 5$
- (C) $x^2 + 2x + 7$
- (D) $x^2 + 2x - 3$
- (E) $x^2 - 2x + 5$

Open Ended Questions (FRQ)

Q9. Let $f(x) = x^2 - 3$ and $g(x) = x - 2$. Find $(f/g)(x)$ and determine its domain.

Q10. The growth rate of a plant is given by $G(x) = x + 1$ and the growth rate of another plant is given by $H(x) = 2x - 1$. Find $(G \cdot H)(x)$ and simplify the resulting expression.

Chef's Tips

- Use a ruler to draw diagrams
- Check units in word problems
- Show all working for marks